

# MATH110

## Course Summary

Course: MATH110      Title: College Algebra

Length of Course: 8

Prerequisites: None

Credit Hours: 3

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## Description

**Course Description:** This course investigates the concepts of college algebra. The course covers the concepts of algebra, graphing and solution of linear and quadratic equations, inequalities, and the solution of systems of linear equations. The course is organized into four distinct parts. The first part of the course covers the basic concepts involved in graphing points and linear equations. The second part of the course investigates the solution and graphing of inequalities and systems of linear equations. The third part of the course concentrates on the manipulation and use of exponential expressions and radicals. The final part of the course considers the solution of quadratic equations and their applications. Practical applications are provided throughout the course. There is careful attention to the presentation of concepts that will become important in the study of analytic geometry, trigonometry, and calculus.

## Course Scope:

The course is organized into four distinct parts.

- The first part of the course covers the basic concepts involved in working with linear equations, inequalities, and polynomials.
- The second part of the course investigates the solution and graphing of systems of linear equations and inequalities.
- The third part of the course concentrates on functions and the use of exponential and logarithmic expressions.
- The final part of the course considers the conic sections.

## Objectives

After completing the course, the student should be able to accomplish these Learning Objectives (LO):

- CO1: Identify algebraic functions and conic sections.
  - CO2: Solve algebraic problems using graphs, equations, and inequalities.
  - CO3: Use the rectangular coordinate system to describe a variety of mathematical functions and their characteristics.
  - CO4: Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.
  - CO5: Construct mathematical models to solve application problems.
  - CO6: Use algebra to make informed decisions.
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## Outline

### Week 1:

#### Learning Outcomes

CO 2, 4

- Solve algebraic problems using graphs, equations, and inequalities.
- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.

#### Reading and Resources

LimeSpring Lesson Week 1

#### Assignments

- Lesson: Week 1
- Discussion 1
- Week 1 Homework
- Week 1 Test

#### Weekly Topics

- Fundamentals of Algebra
- Exponents
- Radicals
- Equations
- Inequalities
- Scientific Notation

## Week 2:

### Learning Outcomes

CO 4, 5

- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.
- Construct mathematical models to solve application problems.

### Reading and Resources

LimeSpring Lesson Week 2

### Assignments

- Lesson: Week 2
- Discussion 2
- Week 2 Homework
- Week 2 Test

### Weekly Topics

- Polynomials

## Week 3:

### Learning Outcomes

CO 3, 4

- Use the rectangular coordinate system to describe a variety of mathematical functions and their characteristics.
- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.

### Reading and Resources

LimeSpring Lesson Week 3

### Assignments

- Lesson: Week 3
- Discussion 3
- Week 3 Homework
- Week 3 Test

### Weekly Topics

- Graphing
- The Distance & Midpoint Formulas

### Week 4:

#### Learning Outcomes

CO 1, 2, 3, 4, 5

- Identify algebraic functions and conic sections.
- Solve algebraic problems using graphs, equations, and inequalities.
- Use the rectangular coordinate system to describe a variety of mathematical functions and their characteristics.
- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.
- Construct mathematical models to solve application problems.

#### Reading and Resources

LimeSpring Lesson Week 4

#### Assignments

- Lesson: Week 4
- Discussion 4
- Week 4 Homework
- Week 4 Test

### Weekly Topics

- Systems of Equations
- Inequalities

### Week 5:

#### Learning Outcomes

CO 1, 4, 5, 6

- Identify algebraic functions and conic sections.
- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.
- Construct mathematical models to solve application problems.

- Use algebra to make informed decisions.

## Reading and Resources

### LimeSpring Lesson Week 5

## Assignments

- Lesson: Week 5
- Discussion 5
- Week 5 Homework
- Week 5 Test

## Weekly Topics

- Functions
- Function and Function Notation
- Applications of Functions

## Week 6:

### Learning Outcomes

#### CO 1, 2, 3, 4, 5

- Identify algebraic functions and conic sections.
- Solve algebraic problems using graphs, equations, and inequalities.
- Use the rectangular coordinate system to describe a variety of mathematical functions and their characteristics.
- Discuss algebraic concepts and their characteristics, and how they apply to situations in everyday life.

## Reading and Resources

### LimeSpring Lesson Week 6

## Assignments

- Lesson: Week 6
- Discussion 6
- Week 6 Homework
- Week 6 Test

### Weekly Topics

- Exponential and Logarithmic Functions
- Graphs of Exponential and Logarithmic Functions
- Solving Exponential and Logarithmic Equations
- Applications of Functions

### Week 7:

#### Learning Outcomes

CO 1, 2, 3

- Identify algebraic functions and conic sections.
- Solve algebraic problems using graphs, equations, and inequalities.
- Use the rectangular coordinate system to describe a variety of mathematical functions and their characteristics.

#### Reading and Resources

LimeSpring Lesson Week 7

#### Assignments

- Lesson: Week 7
- Discussion 7
- Week 7 Homework
- Week 7 Test

### Weekly Topics

- Conic Sections:
  - Circles, Ellipses, Parabolas and Hyperbolas

### Week 8:

#### Learning Outcomes

CO 6 Use algebra to make informed decisions.

#### Reading and Resources

LimeSpring Lesson Week 8

## Assignments

- Lesson: Week 8
- Discussion 8
- Final Exam

## Weekly Topics

- Course Review
- Final Examination

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## Evaluation

### Discussion Assignments:

The University requires weekly contact from each student. This requirement can be met by taking the Unit Tests and by participating in the Discussions. A total of 16% of the final grade will be based on participation in these. Discussion postings are expected to be written in complete sentences using correct grammar and spelling. Any posting which requires research must be accompanied by a citation of the references used.

### Homework:

Practice problems are assigned at the end of each of LimeSpring's videos. Don't skip around. Be sure to watch each of the videos to learn the material covered in these practice problems. Grades on these are compiled weekly and make up the homework grade. These homework problems are 28% of your final grade but they are an important factor in your success at mastering the subject. Math is not a spectator sport - one learns math by putting the pencil to the paper!

### Workbooks:

Each week's material includes a workbook of problems related to that week's lecture. Workbook grades are not used in the calculation of the course grade; however, completion of the workbook assignment is necessary for the student to access the weekly exam. It is also possible to return to the workbook for additional practice on the material.

### Weekly Tests:

There will be a graded test during each week of the course. Each will be a 25 question online, open-book, open-note exam. You may not consult with any other person while taking the exam. A total of 35% of the final course grade comes from these tests. These assignments will follow each week of the

course and will be problems or questions like those in the lecture and workbook. They are selected to provide the student with hands on experience in applying the techniques and models being discussed.

### Final Exam:

The final exam will count as 25% of the final grade. It will also be a 25 question online, open-book, open-note exam. You may not consult with any other person while taking the exam. This examination will be based on all material covered during the semester. The questions will require computations and application of the material covered during the semester. Please coordinate with the professor for any special arrangements. Unless the professor approves alternate arrangements, students should plan to take the final examination during the last week of the course. You will not need a proctor to take this exam.

### Grading

| <i>Task</i>               | <i>Grade %</i> |
|---------------------------|----------------|
| <b>Discussions</b>        | <b>16%</b>     |
| <b>Homework/Quizzes</b>   | <b>24%</b>     |
| <b>Weekly Tests/Exams</b> | <b>35%</b>     |
| <b>Final Examination</b>  | <b>25%</b>     |

## Materials

**Book Title:** Various resources from Trefry Library and/or the Open Web are used. Links provided inside the classroom.

All required readings are located in the Reading and Resources tab under the Lessons tab.

### Course Resources

Video lessons are presented by LimeSpring – access to which is provided within the classroom. LimeSpring also provides the homework and weekly tests as well as a workbook designed to give students the necessary practice to be successful with the material.

Students will need a calculator to successfully complete this course. The calculator should include a memory and square root function. At the student's discretion, a scientific calculator capable of performing statistical functions or a computer spreadsheet program like Microsoft Excel may be used. Students may make use of the above for all graded assignments during the course.

### Supplementary Materials

The lessons contain links to online supplementary materials for this class. You may click on the links in the lessons directly to view them. In addition to these, the following public domain web sites are useful. Please abide by the university's academic honesty policy when using Internet sources as well. Note web site addresses are subject to change.

## Course Guidelines

### Citation and Reference Style

Students will follow the APA Format as the sole citation and reference style used in written work



submitted as part of coursework to the University. Assignments completed in a narrative essay or composition format must follow the citation style cited in the APA Format.

### Late Assignments

The University encourages all work to be completed according to the course schedule. The University Late Work Policy can be found in the [Student Handbook here](#).

### Tutoring

Tutor.com offers online homework help and learning resources by connecting students to certified tutors for one-on-one help. AMU and APU students are eligible for 10 free hours\* of tutoring provided by APUS. Tutors are available 24/7 unless otherwise noted. Tutor.com also has a SkillCenter Resource Library offering educational resources, worksheets, videos, websites and career help. Accessing these resources does not count against tutoring hours and is also available 24/7. Please visit the APUS Library and search for 'Tutor' to create an account.

### Turn It In

Faculty may require assignments be submitted to Turnitin.com. Turnitin.com will analyze a paper and report instances of potential plagiarism for the student to edit before submitting it for a grade. In some cases, professors may require students to use Turnitin.com. This is automatically processed through the Assignments area of the course.

### Academic Dishonesty

Academic Dishonesty incorporates more than plagiarism, which is using the work of others without citation. Academic dishonesty includes any use of content purchased or retrieved from web services such as CourseHero.com. Additionally, allowing your work to be placed on such web services is academic dishonesty, as it is enabling the dishonesty of others. The copy and pasting of content from any web page, without citation as a direct quote, is academic dishonesty. When in doubt, do not copy/paste, and always cite.

### Submission Guidelines

Some assignments may have very specific requirements for formatting (such as font, margins, etc.) and submission file type (such as .docx, .pdf, etc.) See the assignment instructions for details. In general, standard file types such as those associated with Microsoft Office are preferred, unless otherwise specified.

### Communicating on the Discussion Forums

Forums are the heart of the interaction in this course. The more engaged and lively the exchanges, the more interesting and fun the course will be. Only substantive comments will receive credit. Although there is a final posting time after which the instructor will grade comments, it is not sufficient to wait until the last day to contribute your comments/questions on the forum. The purpose of the forums is to actively participate in an on-going discussion about the assigned content.

“Substantive” means comments that contribute something new and hopefully important to the discussion. Thus, a message that simply says “I agree” is not substantive. A substantive comment contributes a new idea or perspective, a good follow-up question to a point made, offers a response to a question, provides an example or illustration of a key point, points out an inconsistency in an argument, etc.

As a class, if we run into conflicting viewpoints, we must respect each individual's own opinion. Hateful and hurtful comments towards other individuals, students, groups, peoples, and/or societies will not be tolerated.

## Netiquette

Online universities promote the advancement of knowledge through positive and constructive debate, both inside and outside the classroom. Forums on the Internet, however, can occasionally degenerate into needless insults and flaming. Such activity and the loss of good manners are not acceptable in a university setting. Basic academic rules of good behavior and proper Netiquette must persist.

Remember that you are in a place for the rewards and excitement of learning, which does not include descent to personal attacks or student attempts to stifle the learning of others.

- Humor Note: Despite the best of intentions, jokes and especially satire can easily get lost or taken seriously. If you feel the need for humor, you may wish to add emoticons to help alert your readers: ;-), :), .

## Disclaimer Statement

Course content may vary from the outline to meet the needs of this particular group.

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## Communications

### Student Communication

To reach the instructor, please communicate through the MyClassroom email function accessible from the Classlist of the Course Tools menu, where the instructor and students email addresses are listed, or via the Office 365 tool on the Course homepage.

- In emails to instructors, it's important to note the specific course in which you are enrolled. The name of the course is at the top center of all pages.
- Students and instructors communicate in Discussion posts and other learning activities.
- All interactions should follow APUS guidelines, as noted in the [Student Handbook](#), and maintain a professional, courteous tone.
- Students should review writing for spelling and grammar.
- [Tips on Using the Office 365 Email Tool](#)

### Instructor Communication

The instructor will post announcements on communications preferences involving email and Instant Messaging and any changes in the class schedule or activities.

- Instructors will periodically post information on the expectations of students and will provide feedback on assignments, Discussion posts, quizzes, and exams.
- Instructors will generally acknowledge student communications within 24 hours and respond within 48 hours, except in unusual circumstances (e.g., illness).
- The APUS standard for grading of all assessments (assignments, Discussions, quizzes, exams) is five days or fewer from the due date.
- Final course grades are submitted by faculty no later than seven days after the end date of the course or the end of the extension period.

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## University Policies

Consult the [Student Handbook](#) for processes and policies at APUS. Notable policies:

- [Drop/Withdrawal Policy](#)
- [Extension Requests](#)
- [Academic Probation](#)
- [Appeals](#)
- [Academic Dishonesty / Plagiarism](#)
- [Disability Accommodations](#)
- [Student Deadlines](#)
- [Video Conference Policy](#)

## Mission

The [mission of American Public University System](#) is to provide high-quality higher education with emphasis on educating the nation's military and public service communities by offering respected, relevant, accessible, affordable, and student-focused online programs that prepare students for service and leadership in a diverse, global society.

## Minimum Technology Requirements

- Please consult the catalog for the minimum hardware and software required for [undergraduate](#) and [graduate](#) courses.
- Although students are encouraged to use the [Pulse mobile app](#) with any course, please note that not all course work can be completed via a mobile device.

## Disclaimers

- Please note that course content – and, thus, the syllabus – may change between when a student registers for a course and when the course starts.
- Course content may vary from the syllabus' schedule to meet the needs of a particular group.